Danté M. Herrera

dante.herrera.1999@gmail.com danteherrera.com github.com/danteherrera1999

Awards

LANL LAAP Award (2024)
LANL SPOT Awards (2023, 2024)
Dean's List (5+ Terms)
National Hispanic Recognition Scholar

Certifications

LANL Radiation Worker II
LANL Energized Electrical Worker
LANL Pressurized Systems Worker
Active DOE Q Clearance

Skills

Software
Python
LabVIEW
MATLAB / Simulink
C/C++ (Microcontrollers)
SolidWorks, NX
HTML/CSS/JS, PHP, SQL
MS Office / LaTeX

Engineering
PXIe/cRIO/cDAQ systems
Oscilloscopes/DMMs/Signal Generators
RS-232/485, I2C, UART Interfaces
Sensor Integration
Embedded Controls
Signal Conditioning
Automation & Controls
Vacuum & Gas Systems

Collaboration
Technical Communication
Team Leadership
Public Speaking
Mentoring

Coursework

Physics/Biophysics
Quantum Physics
Thermo & Stat Mech
Advanced Astrophysical Research
Advanced Biophysics Lab
Extragalactic Astronomy
Organic & Physical Chemistry

Aerospace / Application
Computational Fluid Dynamics
Numerical Simulation
Heat Transfer & Viscous Flows
Active Controls for Aerospace Vehicles
Aerothermo & Propulsion Systems
High-Speed Aerodynamics

Education

Arizona State University May 2024 – May 2026 (Expected)
B.S. Biophysics, *Summa Cum Laude* GPA: 4.00

Texas A&M University Aug 2018 – May 2022
B.S. Aerospace Engineering, Astrophysics Minor, *Magna Cum Laude* GPA: 3.79

Research & Engineering Experience

Los Alamos National Laboratory Jul 2022 – Present
R&D Engineer II | Gas Transfer Systems | Los Alamos, NM

- Lead developer for next-generation Gas Transfer Operation Station (GaTOS), integrating LabVIEW, embedded controllers, custom diagnostics hardware, and high-reliability monitoring for multi-million dollar experiments.
- Developed the Valve Surveillance Test Stand (VSTS), a PXIe- based DAQ platform integrating accelerometers, analog/serial pressure transducers, thermocouples, RTDs, vacuum gauges, and custom diagnostic channels in support of a critical NNSA hardware surveillance program.
- Supported major WETF plutonium and tritium experiments by designing DAQ packages, diagnostics integration, and automated control logic for high-consequence tests.
- Built Python automation pipelines for calibration, data reduction, fault detection, and system verification/validation workflows.
- Modernized legacy DAQ hardware by developing modular signal conditioning boards, improving noise performance and enabling the use of modern sensors.
- Designed high-pressure gas-transfer architectures including valve networks, pneumatic control systems, purge paths, relief systems, pump infrastructure, and power distribution / diagnostic sensor networks.
- Collaborated with diagnostics, physics modeling, mechanical design, and operations teams to execute experimental campaigns.
- Trained new engineers and students on DAQ systems, LabVIEW / Python design practices, and experiment instrumentation workflows.

SpaceX Sep 2021 - Dec 2021
Starship Launch Intern | Orbital Launch Pad Fluids | Boca Chica, TX

- Developed Python based software tools for designing and monitoring fluid systems with real-time (stage-0) launch pad sensor data.
- Designed, installed, and operated cryogenic and pneumatic transfer systems supporting Starship integrated testing.
- Coordinated daily with fabrication, welding, and fluids teams to accelerate build-up and readiness of Orbital Launch Pad infrastructure.

TAMU Sounding Rocketry Team Jan 2019 – Sep 2021
Propulsion Team | Hybrid Engine Development | College Station, TX

- Designed hybrid engine combustion chamber, fuel grain geometry, and oxidizer routing using SolidWorks.
- Developed a 1/8th scale R&D hybrid rocket motor enabling additive and grain-geometry testing with 15+ successful firings.
- Designed, built, and characterized a full-scale hybrid sounding rocket engine producing $> 900\ lbf$ average thrust.

Texas A&M Aerospace Engineering Dept. Aug 2018 – May 2021
Undergraduate Researcher | Hybrid Propulsion | College Station, TX

- Awarded 1st Place, undergraduate general engineering poster section, TAMU Student Research Week 2021.
- Developed an HTPB/N_2O regression-rate model using heat transfer, mass flow, and convection models.
- Designed high temperature combustion-chamber measurement and calibration techniques.